CASE STUDY 1

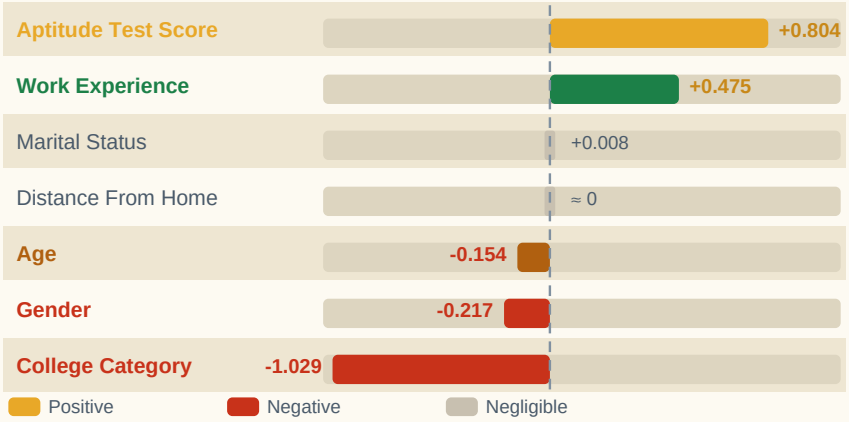
Hiring by Evidence, Not Instinct.

Logistic Regression · 150 Candidates · 7 Predictors

Building a transparent, scalable hiring algorithm for VP-HR Ms. Sarika at Telco India

88%	0.81	2.23×
Model Accuracy	AUC-ROC Score	Aptitude Odds Lift

WHAT ACTUALLY DRIVES CANDIDATE SELECTION? — LOGISTIC REGRESSION COEFFICIENTS (β)



SELECTED

Avg. Aptitude 5.70

Avg. Experience 2.33 yrs

REJECTED

Avg. Aptitude 4.00

Avg. Experience 1.50 yrs

WHAT EACH VARIABLE MEANS FOR YOUR HIRING PROCESS

PRIORITISE

Aptitude Test

Strongest signal (β=+0.80). A +1 SD improvement more than doubles selection odds. Set a minimum score threshold before any candidate advances to interview stage.

WEIGHT HEAVILY

Work Experience

Second strongest (β=+0.48). Selected candidates averaged 2.33 yrs vs 1.50 yrs — a 55% gap. Assign explicit rubric points to prior roles in CV screening.

AUDIT FIRST

College Category

Strongest negative (β=−1.03). May reflect historical bias. Validate against actual job performance before retaining as a criterion. Could unfairly exclude talent.

REMOVE ENTIRELY

Age & Gender

Near-zero predictive value yet carry real legal compliance risk. Removing both protects the organisation from discrimination liability without reducing accuracy.

STRATEGIC RECOMMENDATIONS

VERDICT: DEPLOY AS DECISION-SUPPORT TOOL

01 IMMEDIATE

Gate on Aptitude First

Make the aptitude test the first substantive filter. Set the score cutoff empirically from previously selected candidates. No candidate below the threshold advances without exceptional justification from the hiring panel.

02 SHORT-TERM

Reward Demonstrated Experience

Assign explicit rubric weight to work experience in CV screening and interview scorecards. For technical and senior roles, relevant prior experience is the best available proxy for on-the-ground productivity and readiness.

03 AUDIT FIRST

Validate the College Tier Criterion

Run a performance analysis before encoding this. Do Tier 1 hires actually outperform others on measurable outcomes? If evidence is mixed, downweight or remove this criterion entirely to prevent perpetuating historical bias.

04 NEXT CYCLE

Address Class Imbalance in Future Iterations

With only 18% of candidates selected, the model skews conservative — it rejects well but misses some strong candidates. Apply SMOTE or adjusted class weights in future versions to improve sensitivity without sacrificing specificity.

CASE STUDY 5

From 26 Survey Items to 5 Actionable Dimensions.

Exploratory Factor Analysis (PCA) · 220 Employees · 26-Item Likert Survey

Uncovering latent engagement structure across 7 cities — CHRO Ms. Aniketna Dutta · Lead: Sanket Sharma

73.5%	0.948	41.9%
Variance Explained	Cronbach α F1	Factor 1 Variance

THE 5 ENGAGEMENT DIMENSIONS IDENTIFIED THROUGH FACTOR ANALYSIS

F1 — Organizational Commitment & Recognition
14 items
α = 0.948
Score 3.81
✔ Reliable

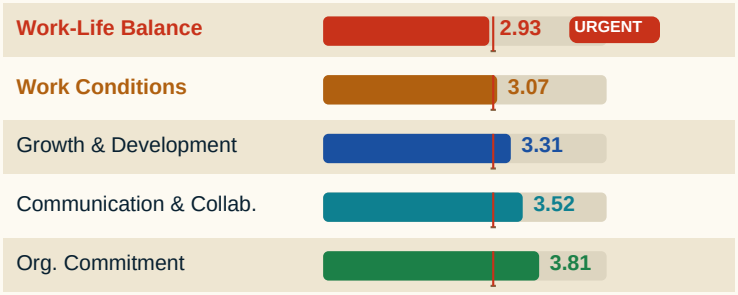
F2 — Work Conditions & Autonomy
2 items
α = 0.605
Score 3.07
■ Thin

F3 — Communication & Collaboration
4 items
α = 0.813
Score 3.52
✔ Reliable

F4 — Work-Life Balance
2 items
α = 0.609
Score 2.93
■ Urgent

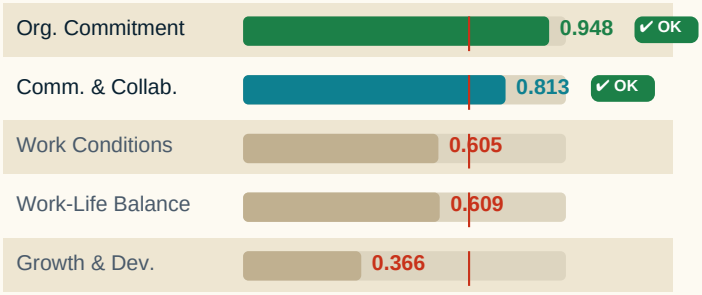
F5 — Growth & Development
2 items
α = 0.366
Score 3.31
✗ Weak α

MEAN ENGAGEMENT SCORE (1–5 LIKERT SCALE)



Neutral (3.0) — below = disengaged

INTERNAL RELIABILITY — CRONBACH'S ALPHA



Threshold α=0.70 — above = reliable

WHAT SHOULD ABL LTD DO? — STRATEGIC RECOMMENDATIONS

CHRO: MS. ANIKETNA DUTTA · LEAD: SANKET SHARMA

01 — IMMEDIATE

Fix Work-Life Balance Now

Score 2.93 — the only dimension below neutral. Run office-level diagnostics immediately. Introduce shift transparency, structured workload reviews, and a compensatory leave policy. Audit UP offices first given Aniketna's direct remit.

02 — SHORT-TERM

Improve Work Conditions & Autonomy

Score 3.07 — just above neutral. Invite floor-level employees into process-improvement discussions. Give team leads more discretionary authority. Even modest autonomy gains produce outsized engagement impact in constrained manufacturing environments.

03 — SHORT-TERM

Strengthen Cross-Office Communication

Score 3.52 across 7 dispersed locations. Deploy digital collaboration tools, establish regular cross-city touchpoints, and create shared project workstreams mixing employees from Kanpur, Lucknow, Varanasi, Agra, and Gujarat offices.

04 — SUSTAIN

Protect Organizational Commitment

Score 3.81, α=0.948 — the strongest asset. Employees have genuine, consistent belonging. Maintain through recognition practices, transparent leadership communication, and deliberate culture-building that includes all 7 cities, not just HQ.

05 — NEXT CYCLE

Redesign the Survey Instrument

Factors 2, 4 & 5 have only 2 items each — too thin for reliable measurement. Add 3–4 items per weak factor. Each dimension needs 4–5 items minimum to achieve psychometric validity before the next annual survey cycle.

06 — NEXT CYCLE

Segment Scores by Region

220 respondents treated as one population. Break factor scores by office to reveal UP vs Gujarat differences and build a region-specific engagement map — enabling Aniketna to allocate intervention resources where they are most needed.